Garrett G. Wen

Ph.D. Candidate
Department of Statistics and Data Science
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Google Scholar

Education

- **Yale University** — **New Haven, CT, USA**
  Ph.D. candidate, Department of Statistics and Data Science. — 09/2022 – Present
  Ph.D. Advisors: Prof. Theodor Misiakiewicz and Prof. Zhou Fan.
  Expected graduation / degree conferral: May 24, 2027.
  M.A. in Statistics (en route), conferred 12/2025.

- **Peking University** — **Beijing, China**
  B.Sc. in Statistics, School of Mathematical Sciences. — 09/2018 – 06/2022

Academic & Research Experience

- **Flatiron Institute, Center for Computational Mathematics (CCM)** — **New York, NY**
  Summer Research Associate. — 06/01/2026 – 08/14/2026
  Advisors: Dr. Alberto Bietti, Dr. Nikhil Ghosh, and Dr. Denny Wu.

- **The Wharton School, University of Pennsylvania** — **Philadelphia, PA**
  Visiting Researcher, Department of Statistics and Data Science. — 05/2023 – 07/2023
  07/2024 – 08/2024; 05/2025 – 07/2025
  Remote Research Intern, Department of Statistics and Data Science. — 03/2021 – 02/2022
  Research on deep learning theory and peer-review mechanisms.
  Advisor: Prof. Weijie Su.

- **MEGVII**, Beijing, China — 02/2022 – 08/2022
  Research Intern, Automated Theorem Proving with Deep Learning.
  Advisor: Dr. Shuchang Zhou.

- **Peking University**, Beijing, China — 08/2019 – 02/2022
  Undergraduate Researcher, School of Mathematical Sciences.
  Advisor: Prof. Zhihua Zhang.

Summer Schools

- **Statistical Physics & Machine Learning: Moving Forward**, Cargèse, France — 08/2025

- **Princeton Machine Learning Theory Summer School**, Princeton, NJ — 06/2023

Research Interests

- Deep learning theory, non-convex optimization, and high-dimensional learning dynamics
- High-dimensional statistics, probability theory, and random matrix theory
- Generative AI, including language-model watermarking and synthetic data
- Mechanism design for scientific evaluation and peer review

Awards & Honors

- Yale Graduate Student Assembly Conference Travel Fellowship 10/2025
- Award for Academic Excellence, Peking University 2019, 2020, 2021, 2022
- **First Prize**, 11th Chinese Mathematics Competition for College Students, National Final (Senior Division) 04/2021
- Guangzhou Pharmaceutical Wanglaoji Scholarship, Peking University 2021
- CreditEase Internet Finance Scholarship, Peking University 2020

Teaching Experience

- **Teaching Fellow (TF) at Yale University** Fall 2024 – Spring 2026
  - S&DS 312/S&DS 612: Linear Models Fall 2024/Fall 2025
    Instructor: Prof. Zongming Ma.
  - S&DS 361/AMTH 361/S&DS 661: Data Analysis Spring 2025
    Instructor: Prof. Brian Macdonald.
  - S&DS 6590: Mathematics of Deep Learning Spring 2026
    Instructor: Prof. Theodor Misiakiewicz. Delivered one course lecture on benign overfitting in February 2026.

Publications

$*$ denotes equal contribution; in (J1), the two co-first authors are listed with Gang Wen first.

Conference and Workshop Papers:

(C2) "Fast Learning Rate Transfer for Gradient Descent in Sketched Linear Regression."
Garrett Wen, Alberto Bietti, Nikhil Ghosh, Theodor Misiakiewicz, Denny Wu.
ICML Workshop on High-Dimensional Learning Dynamics (HiLD), **spotlight**, 2026.

(C1) "PeriodicSketch: Finding periodic items in data streams."
Zhuochen Fan, Yinda Zhang, Tong Yang, Mingyi Yan, Gang Wen, Yuhan Wu, Hongze Li, Bin Cui.
2022 IEEE 38th International Conference on Data Engineering (ICDE), pp. 96–109, 2022.

Journal Papers:

(J4) "Recommending Best Paper Awards via the Isotonic Mechanism."
Garrett G. Wen, Buxin Su, Natalie Collina, Zhun Deng, Weijie Su.
Accepted for publication in *Journal of Machine Learning Research*, 2026.

(J3) "Optimal Estimation of Watermark Proportions in Hybrid AI-Human Texts."
Xiang Li, Garrett Wen, Weiqing He, Jiayuan Wu, Qi Long, Weijie J. Su.
Minor revision at *Journal of the American Statistical Association*, 2026.

(J2) "Self-rankings as a predictor of scientific impact beyond peer review."
Buxin Su, Natalie Collina, Garrett Wen, Didong Li, Kyunghyun Cho, Jianqing Fan, Bingxin Zhao, Weijie Su.
Nature Computational Science, 2026. DOI: 10.1038/s43588-026-01039-0.

(J1) "On the Evolutionary of Bloom Filter False Positives—An Information Theoretical Approach to Optimizing Bloom Filter Parameters."
Gang Wen*, Zhuochen Fan*, Zhipeng Huang, Yang Zhou, Qiaobin Fu, Tong Yang, Alex X. Liu, Bin Cui.
IEEE Transactions on Knowledge and Data Engineering, 35(7), pp. 7316–7327, 2023.

Preprints

** denotes alphabetical order.

(P4) "The sharp SAT/UNSAT phase transition in random ellipsoid fitting."
Theodor Misiakiewicz**, Garrett G. Wen**.
Submitted to *The Annals of Probability*; arXiv:2608.10184, 2026.

(P3) "Dynamical mean-field limit and replica-symmetric free energy for the orthogonally-invariant SK model."
Zhou Fan**, Theodor Misiakiewicz**, Leda Wang**, Garrett G. Wen**.
Submitted to *The Annals of Probability*; arXiv:2607.10102, 2026.

(P2) "When does Gaussian equivalence fail and how to fix it: Non-universal behavior of random features with quadratic scaling."
Garrett G. Wen, Hong Hu, Yue M. Lu, Zhou Fan, Theodor Misiakiewicz.
Preprint arXiv:2512.03325, 2025.

(P1) "Optimal Detection for Language Watermarks with Pseudorandom Collision."
T. Tony Cai**, Xiang Li**, Qi Long**, Weijie J. Su**, Garrett G. Wen**.
Submitted to *IEEE Transactions on Information Theory*; arXiv:2510.22007, 2025.

Selected Talks and Presentations

(T3) *International Conference on Statistics and Data Science (ICSDS)*, IMS, Seville — 12/2025

(T2) "High-Dimensional Limit Theorems for SGD: Effective Dynamics and Critical Scaling," *Machine Learning Theory Reading Group*, Yale University — 09/2024

(T1) "The Isotonic Mechanism for Recommending Best Paper Awards," poster presentation, *Workshop on Incentives in Academia (WINA-24)*, New Haven — 07/2024

Service

- **Reviewer:** Neural Information Processing Systems (NeurIPS), International Conference on Learning Representations (ICLR), ICML Workshop on High-Dimensional Learning Dynamics (HiLD), Journal of Machine Learning Research (JMLR), and Journal of the Royal Statistical Society: Series B (Statistical Methodology).

Technical Skills

- **Programming:** Python, R, MATLAB, SQL, C/C++, C#, Java, CUDA.
- **Tools:** PyTorch, TensorFlow, Git, LaTeX, Linux/Ubuntu, Vim, Unity, Origin.

References

- **Prof. Theodor Misiakiewicz**, Department of Statistics and Data Science, Yale University.
  theodor.misiakiewicz@yale.edu
- **Prof. Zhou Fan**, Department of Statistics and Data Science, Yale University.
  zhou.fan@yale.edu
- **Prof. Weijie Su**, Department of Statistics and Data Science, The Wharton School, University of Pennsylvania.
  suw@wharton.upenn.edu
- **Dr. Alberto Bietti**, Center for Computational Mathematics, Flatiron Institute.
  abietti@flatironinstitute.org
- **Dr. Denny Wu**, Faculty Fellow, Center for Data Science, New York University; Center for Computational Mathematics, Flatiron Institute.
  dennywu@nyu.edu